



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
COMPUTER SCIENCE AND ENGINEERING PROGRAMME
CSA51 – CRYPTOGRAPHY AND NETWORK SECURITY



Assignment1

Question1:

- **Encrypt** the message "DEFEND THE WALL" using Caesar cipher with key = 3.
- **Decrypt** the message "KHOOR ZRUOG" using Caesar cipher with key = 3.
- A Caesar cipher is used to encrypt the word "SECRET" to "VHFUHW".
Determine the key used, and decrypt "WKL V D WHVW".

Question2.

- Using the substitution key:
A→Q, B→W, C→E, D→R, E→T, F→Y, ..., Z→M
Encrypt the word "FIRE".
- Given this cipher text "UJHHD QEB NRFZH YOLTK",
and knowing it is encrypted with a **monoalphabetic substitution**,
Find the plaintext using frequency analysis (assume it's English).

Question3.

- **Encrypt** the message "ATTACKATPALACE" using the keyword "LEMON".
- **Decrypt** the cipher text "LXFOPVEFRNHR" using the keyword "LEMON".
- Given the encrypted message "QNXEPKMAEGTU" using the Vigenère cipher and keyword "KEY",
Find the original message.

Question4.

Given a 2x2 key matrix:

- $$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Encrypt the message "HI" using the Hill cipher.
(Use A=0, B=1, ..., Z=25.)
- If the message "PO" was encrypted to "QC" using the above Hill cipher matrix,
Verify the result by showing the matrix multiplication.

Question5.

- **Encrypt** the message "MEETMEAFTERLUNCH" using the keyword "ZEBRAS".
- **Decrypt** the message "EANMTEFCUHEHETREL" using the keyword "BOARD".

Question6.

- Using the one-time pad key **XMCKL**, encrypt the message **"HELLO"**.
- Explain why the message **"ATTACKNOW"** encrypted using OTP with key **"QWERTYUIO"** is **unbreakable** under ideal conditions.

Question7.

- **Encrypt** the message:
 - ◆ Plaintext: "WE ARE DISCOVERED"
 - ◆ Keyword: "ZEBRA"

Steps expected:

- Arrange the message in rows under the keyword.
 - Determine column order based on alphabetical sorting of keyword letters.
 - Read columns in sorted order to form ciphertext.
-
- **Decrypt** the message:
 - ◆ Ciphertext: "EVLNEACDTKESEAQROFOJDEECUXWN"
 - ◆ Keyword: "TRIAL"
- Task:**
- Reconstruct the matrix based on the keyword.
 - Rearrange the columns to the original order.
 - Read row-wise to get the plaintext.
-
- The message **"INFORMATIONSECURITY"** is encrypted using columnar transposition with keyword **"HACK"**.
 1. Encrypt the message.
 2. Explain how many rows are needed.
 3. Add necessary padding characters (like 'X') if needed.

Question8.

You are given the following cipher text encrypted using a **monoalphabetic substitution cipher**:

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUZUETUCG

Assume it is an English sentence.

Task:

1. Perform **frequency analysis** of letters.
2. Use standard English letter frequencies (e.g., E, T, A, O are most common) to **hypothesize probable mappings**.

3. Attempt to **partially decrypt** the message.
4. Provide the best guess for the plaintext message (even partial guesses allowed).

Question9.

You intercepted a message encrypted with a monoalphabetic substitution cipher:

Ciphertext: BJJW QJW YMJW NX F XNRUQJWX

You know the original plaintext starts with:
"THIS IS THE".

Task:

1. Use the known-plaintext portion to **find substitution mappings**.
2. **Complete the decryption** of the full message.
3. Validate if any letters repeat and adjust your substitution table if necessary.

Question10.

You are given the cipher text below, encrypted using the Vigenère cipher with an unknown short key (2-5 letters long):

Ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Try to recover the key and decrypt the full message.